

Hobbsleigh

August 2002 (3 hours)

Total marks (120)

Attempt Questions 1-8

All questions are of equal value

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

QUESTION 1 (15 marks) Use a SEPARATE writing booklet.

Marks

(a) Find $\int \frac{dx}{x^2 + 2x + 5}$.

2

(b) (i) Find real numbers A, B and C such that

$$\frac{3x+7}{(x+1)(x+2)(x+3)} = \frac{A}{x+1} + \frac{B}{x+2} + \frac{C}{x+3}$$

2

(ii) Hence evaluate $\int_0^1 \frac{3x+7}{(x+1)(x+2)(x+3)} dx$

2

(c) Show $\int e^{ax} \sin 3x dx = \frac{e^{ax}}{a^2+9} [a \sin 3x - 3 \cos 3x]$ and hence evaluate $\int_0^{2\pi} e^{2x} \sin 3x dx$.

4

(d) $I_m = \int x^m e^x dx$

(i) Show that $I_m = x^m e^x - m I_{m-1}$

2

(ii) Find the value of $\int_1^2 x^2 e^x dx$

3

End of Question 1

Marks

QUESTION 2 (15 marks) Use a SEPARATE writing booklet.

(a) The complex number z is given by $z = 1 + \frac{1+i}{1-i}$. Find

(i) $\bar{z}$, giving your answer in the form $x+iy$ where x and y are real.

2

(ii) iz

1

(b) Find u and v if $(u-iv)^2 = -21-20i$.

4

(c) If $z_1 = 1+3i$ and $z_2 = 1-3i$, find $\left| \frac{z_1^{10}}{z_2^9} \right|$.

3

(d) Shade the region in the complex plane for which $2 \leq z + \bar{z} \leq 6$.

2

(e) In the Argand diagram $P(z)$ is a point in the first quadrant of the circle $|z|=2$. If $\arg z = \theta$, find, in terms of θ , expressions for

(i) $\arg z^2$

1

(ii) $\arg(z-2)$

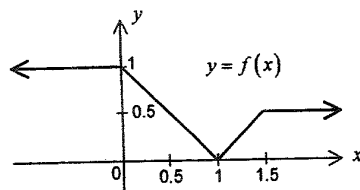
2

End of Question 2

QUESTION 3 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) The diagram below is a sketch of the function $y = f(x)$.



On separate diagrams sketch

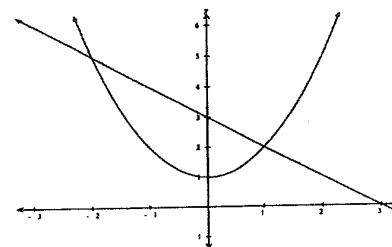
- | | | |
|-------|--|---|
| (i) | $y = f(-x)$ | 2 |
| (ii) | $y = \frac{1}{f(x)}$ | 2 |
| (iii) | $ y = f(x)$ | 2 |
| | | |
| (b) | (i) Express $\frac{x^2-8}{x^2-4}$ in the form $c + \frac{d}{x^2-4}$ where c and d are integers. | 1 |
| | (ii) Draw a neat sketch of $y = \frac{x^2-8}{x^2-4}$. Clearly indicate the intercepts with the coordinate axes and the position and equation of all asymptotes. | 4 |
| | (iii) Shade the region R described by $0 \leq y \leq \frac{x^2-8}{x^2-4}$ and $3 \leq x \leq 4$. | 1 |
| | (iv) Show that the area of $R = 2 \left(1 + \log_e \frac{3}{5} \right)$. | 3 |

End of Question 3

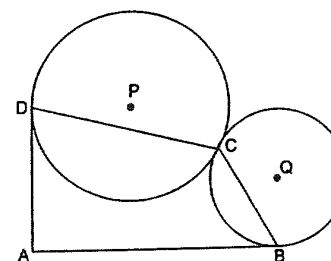
QUESTION 4 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) Show that $2 - \sqrt{3}$ is a zero of the polynomial $a(x) = x^3 - 15x + 4$.
Hence reduce $a(x)$ to irreducible factors over the real field. 3
- (b) Factorise $Q(x) = x^6 - 3x^2 + 2$ over the field of complex numbers, given that it has two double roots. 4
- (c) The area bounded by the curve $y = x^2 + 1$ and the line $y = 3 - x$ is rotated about the x -axis to form a solid.



- (i) By considering slices perpendicular to the x -axis show that the area of one slice is given by $A = \pi(8 - 6x - x^2 - x^4)$. 2
- (ii) Hence find the volume of the solid formed. 2
- (d) Two circles touch each other at C . AD is a tangent to the circle with centre P and touches the circle at D . AB is a tangent to the circle with centre Q and touches the circle at B .



NOT TO SCALE

Copy the diagram into your answer booklet.

Prove that $\angle BCD = 180^\circ - \frac{1}{2} \angle BAD$. 4

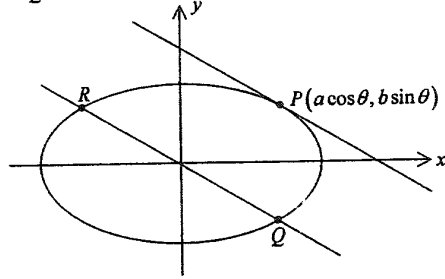
End of Question 4

QUESTION 5 (15 marks) Use a SEPARATE writing booklet.

- (a) The velocity v cm/s of a particle moving in a straight line is given by $v^2 = 48 + 16x - 4x^2$ where x cm is the displacement of the particle from a fixed point.
- (i) Show that the particle is moving with simple harmonic motion. Hence write down the centre of motion. 3
- (ii) What is the amplitude of the motion? 2
- (iii) If initially the particle is at one of the extreme points, how far will it travel in the first $\frac{3\pi}{4}$ seconds? 2
- (iv) Where is the particle when $t = \frac{3\pi}{4}$ seconds? 1
- (b) A torus is generated by revolving the region $x^2 + y^2 \leq 4$ about the line $x = 5$.
- (i) By using the method of cylindrical shells show that the volume of one shell is given by $\Delta V = 4\pi(5-x)\sqrt{4-x^2}\Delta x$. 2
- (ii) Hence find the volume of the torus. 2
- (c) Solve the equation $4x^3 - 12x^2 + 11x - 3 = 0$ given that the roots form an arithmetic series. 3

End of Question 5

QUESTION 6 (15 marks) Use a SEPARATE writing booklet.

- (a) Prove by mathematical induction $3^n - 1 \geq 2n$ where n is a positive integer. 3
- (b) $P(a\cos\theta, b\sin\theta)$ is a point on the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with centre O .
A line drawn through O , parallel to the tangent to the ellipse at P , meets the ellipse at Q and R .
- 
- (i) Show that Q and R are the points $(a\sin\theta, -b\cos\theta)$ and $(-a\sin\theta, b\cos\theta)$. 2
- (ii) Prove that the area of $\triangle PQR$ is independent of the position of P . 3
- (c) A solid is built on the ellipse $\frac{x^2}{9} + \frac{y^2}{4} = 1$ such that cross-sections perpendicular to the x -axis are squares with a side in the elliptical base. Find the volume of the solid. 4
- (d) Z and W represent the complex numbers z and w respectively. If $|z| = 2$ and $w = \frac{z+3}{z}$, find the locus of W . 3

End of Question 6

QUESTION 7 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) The tangent to the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ at $P(4\sqrt{2}, 3)$ meets the asymptotes of the hyperbola at A and B .
Show that P is the midpoint of AB . 4
- (b) LMN is an isosceles triangle with $LM = LN$ and P is a point on the bisector of $\angle MLN$. MP produced meets LN at Q and NP produced meets LM at R .
Prove that M, N, Q and R are concyclic. 4
- (c) (i) Show that the equation of the chord joining the points $P\left(2p, \frac{2}{p}\right)$ and $Q\left(2q, \frac{2}{q}\right)$ on the hyperbola $y = \frac{4}{x}$ is $x + pqy = 2(p + q)$. 3
- (ii) By letting q approach p , or otherwise, write down the equation of the tangent to the curve at P . 1
- (iii) If the chord in (i) passes through the point $(2(p + q - 1), 2)$, and T is the point of intersection of the tangents at P and Q , show that the locus of T is $y = x$. 3

End of Question 7

QUESTION 8 (15 marks) Use a SEPARATE writing booklet.

Marks

- (a) Show that $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$.
Use this result to solve the equation $8x^3 - 6x = -1$ for x . 5
- (b) If $a > b$ and $c > d$, prove that $ac + bd > ad + bc$. 2
- (c) If a, b, c, d are positive numbers with $a \geq c + d$ and $b \geq c + d$, prove that $ab \geq ad + bc$. 3
- (d) (i) Show that $\frac{1 + \sin \theta + i \cos \theta}{1 + \sin \theta - i \cos \theta} = \sin \theta + i \cos \theta$ 3
- (ii) Hence prove that
$$\left(\frac{1 + \sin \theta + i \cos \theta}{1 + \sin \theta - i \cos \theta} \right)^n = \cos \left(\frac{n\pi}{2} - n\theta \right) + i \sin \left(\frac{n\pi}{2} - n\theta \right)$$

where n is a positive integer. 2

END OF PAPER

1001

$$\int \frac{dx}{x^2+2x+5} = \int \frac{dx}{(x+1)^2+4}$$

$$= \frac{1}{2} \tan^{-1} \left(\frac{x+1}{2} \right) + C$$

$$3x+7 = A(x+2)(x+3) + B(x+1)(x+3)$$

$$= Ax^2 + 5Ax + 6A + Bx^2 + 4Bx + 3B$$

$$= Ax^2 + 5Ax + 6A + Bx^2 + 4Bx + 3B$$

$$\begin{cases} A+B+C=0 \\ 5A+4B+3C=3 \\ 6A+3B+2C=7 \end{cases} \quad \begin{matrix} A=2, B=-1 \\ C=-1 \end{matrix}$$

$$\frac{x+7}{(x+2)(x+3)} = \frac{2}{x+1} - \frac{1}{x+2} - \frac{1}{x+3}$$

$$\int_0^1 \frac{3x+7}{(x+1)(x+2)(x+3)} dx$$

$$= \left[2\ln|x+1| - \ln|x+2| - \ln|x+3| \right]_0^1$$

$$= 2\ln 2 - \ln 3 - \ln 4$$

$$- (2\ln 1 - \ln 2 - \ln 3)$$

$$= \ln 2$$

$$\int e^{ax} \sin 3x dx$$

$$= -\frac{1}{3} e^{ax} \cos 3x + \frac{a}{3} \int e^{ax} \cos 3x dx$$

$$= -\frac{1}{3} e^{ax} \cos 3x + \frac{a}{3} \left(\frac{1}{3} e^{ax} \sin 3x - \frac{a}{3} \int e^{ax} \sin 3x dx \right)$$

$$\frac{a^2}{9} \int e^{ax} \sin 3x dx = \frac{e^{ax}}{9} (a \sin 3x - 3 \cos 3x)$$

$$\int e^{ax} \sin 3x dx = \frac{e^{ax}}{9+a^2} (a \sin 3x - 3 \cos 3x)$$

$$\int_0^{2\pi} e^{2x} \sin 3x dx = \left[\frac{e^{2x}}{13} (2 \sin 3x - 3 \cos 3x) \right]_0^{2\pi}$$

$$\int_0^{2\pi} e^{-x} \sin 3x dx = \frac{1}{13} (2 \sin 0 - 3 \cos 0)$$

$$= -\frac{e^0}{13} (2 \sin 0 - 3 \cos 0)$$

$$= \frac{3}{13} (1 - e^{4\pi})$$

$$(d) (i) \int x^m e^x dx = x^m e^x - \int m x^{m-1} e^x dx$$

$$= x^m e^x - m \int x^{m-1} e^x dx$$

$$= x^m e^x - m I_{m-1}$$

$$(ii) \int_1^2 x^2 e^x dx = \left[x^2 e^x \right]_1^2 - 2 \int_1^2 x e^x dx$$

$$= 4e^2 - e - 2 \left(\left[x e^x \right]_1^2 - \int_1^2 e^x dx \right)$$

$$= 4e^2 - e - 4e^2 + 2e + 2 \left[e^x \right]_1^2$$

$$= e + 2(e^2 - e)$$

$$= 2e^2 - e$$

Question 2

$$(a) z = 1 + \frac{1+i}{1-i} \times \frac{1+i}{1+i}$$

$$= 1 + i$$

$$(i) \bar{z} = 1 - i$$

$$(ii) iz = -1 + i$$

$$(b) (u - iv)^2 = u^2 - v^2 - 2iuv$$

$$u^2 - v^2 = -21$$

$$2uv = 20 \Rightarrow v = \frac{10}{u}$$

$$u^2 - \frac{100}{u^2} = -21$$

$$u^4 + 21u^2 - 100 = 0$$

$$(u^2 + 25)(u^2 - 4) = 0$$

$$u^2 = -25, u^2 = 4$$

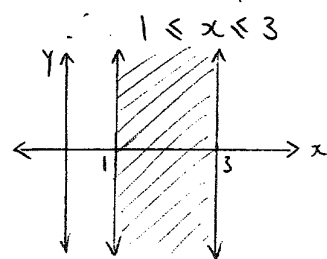
$$u = \pm 5i, \pm 2$$

$$v = \mp 2i, \pm 5$$

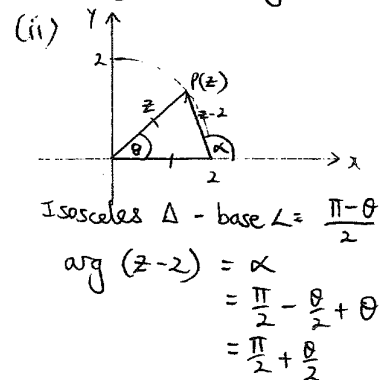
$$|z_2| = |1 - 3i| = \sqrt{10}$$

$$\left| \frac{z_2^{10}}{z_2^9} \right| = \sqrt{10}$$

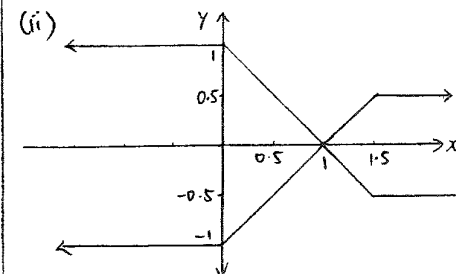
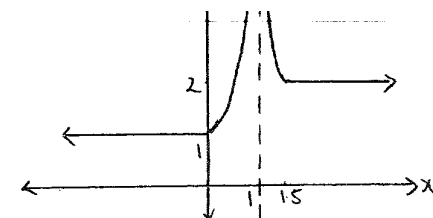
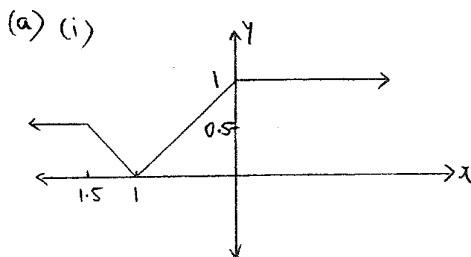
$$(d) z + \bar{z} = x + iy + x - iy = 2x$$



$$(e) (i) \arg z^2 = 2 \arg z = 2\theta$$



Question 3



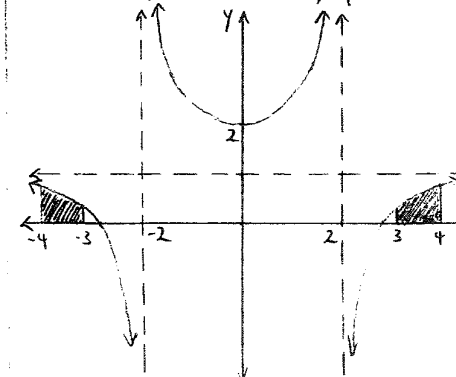
$$(b) (i) \frac{x^2-8}{x^2-4} = \frac{x^2-4}{x^2-4} - \frac{4}{x^2-4}$$

$$= 1 - \frac{4}{x^2-4}$$

$$(ii) \text{intercepts } x = \pm 2\sqrt{2}$$

$$y = 2 \text{ (min + pt.)}$$

$$\text{asymptotes } x = \pm 2, y = 1$$



$$(iv) R = 2 \int_3^4 \left(1 - \frac{1}{x-2} + \frac{1}{x+2} \right) dx$$

$$= 2 \left[x - \ln|x-2| + \ln|x+2| \right]_3^4$$

$$= 2(4 - \ln 2 + \ln 6 - (3 - \ln 1 + 1))$$

$$= 2(1 + \ln \frac{3}{5})$$

tion 4

$$a(2-\sqrt{3}) = (2-\sqrt{3})^3 - 15(2-\sqrt{3}) + 4$$

$$= 8 - 12\sqrt{3} + 18 - 3\sqrt{3} - 30 + 15\sqrt{3} + 4$$

$$= 0$$

$2-\sqrt{3}$ is a zero

conjugate surd $2+\sqrt{3}$ is a zero

$$\chi(x) = (x - (2-\sqrt{3}))(x - (2+\sqrt{3}))(x - b)$$

$$-b(2-\sqrt{3})(2+\sqrt{3}) = 4 \Rightarrow b = -4$$

$$\chi(x) = (x - 2 + \sqrt{3})(x - 2 - \sqrt{3})(x + 4)$$

$$Q(x) = x^6 - 3x^2 + 2$$

$$Q'(x) = 6x^5 - 6x$$

double root $\Rightarrow Q'(x) = 0$

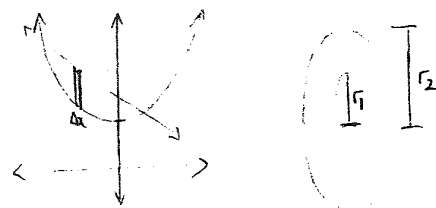
$$6x(x^4 - 1) = 0$$

$$x = 0, \pm 1, \pm i$$

test these $\Rightarrow Q(1) = Q(-1) = 0$

$$Q(x) = (x-1)^2(x+1)^2(x^2+2)$$

$$Q(x) = (x-1)^2(x+1)^2(x-\sqrt{2}i)(x+\sqrt{2}i)$$



$$i) A(x) = \pi(r_2^2 - r_1^2)$$

$$= \pi((3-x)^2 - (x^2+1)^2)$$

$$= \pi(9 - 6x + x^2 - x^4 - 2x^2 - 1)$$

$$= \pi(8 - 6x - x^2 - x^4)$$

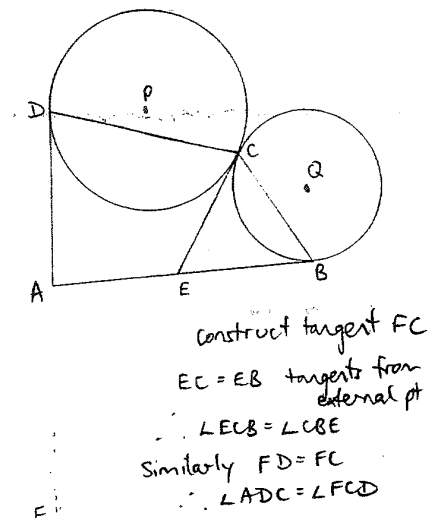
$$ii) \Delta V = \pi(8 - 6x - x^2 - x^4) \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=-2}^6 \pi(8 - 6x - x^2 - x^4) \Delta x$$

$$= \pi \int_{-2}^6 (8 - 6x - x^2 - x^4) dx$$

$$= \pi [8x - 3x^2 - \frac{x^3}{3} - \frac{x^5}{5}]_{-2}^6 = 117\pi \text{ m}^3$$

(d)



Construct tangent FC
 $EC = EB$ tangents from external pt
 $\therefore \angle ECB = \angle CBE$
 Similarly $FD = FC$
 $\therefore \angle ADC = \angle FCD$

$$\angle BCD = 360^\circ - (\angle ABC + \angle ADC + \angle BAD)$$

$$= 360^\circ - (\angle ECB + \angle ECD) - \angle BAD$$

$$= 360^\circ - \angle BCD - \angle BAD$$

$$2\angle BCD = 360^\circ - \angle BAD$$

$$\therefore \angle BCD = 180^\circ - \frac{1}{2}\angle BAD$$

Question 5

$$(a) (i) v^2 = 48 + 16x - 4x^2$$

$$\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} \dot{x}^2 \right)$$

$$= \frac{d}{dx} (24 + 8x - 2x^2)$$

$$= 8 - 4x$$

$= -4(x-2)$
 which is the form of SHM
 with centre $x=2$

$$(ii) \text{ endpts when } v=0$$

$$x^2 - 4x - 12 = 0$$

$$(x-6)(x+2) = 0$$

oscillates between $x=-2$ and $x=6$
 $\therefore$ amplitude 4 cm

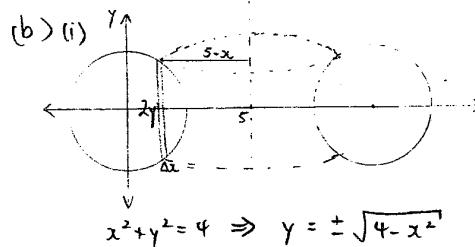
$$(iii) n^2 = 4 \Rightarrow n = 2 \therefore \text{period} = \pi \text{ s}$$

In π s travels 16 cm

$\frac{3\pi}{4}$ is $\frac{3}{4}$ of period

$\therefore$ travels 12 cm

(iv) starts at endpoint $\therefore$ in $\frac{3\pi}{4}$ s
 is at centre of motion $x=2$.



$$\Delta V = 2\pi(5-x)2y \Delta x$$

$$= 4\pi(5-x)\sqrt{4-x^2} \Delta x$$

$$(ii) V = \lim_{\Delta x \rightarrow 0} \sum_{x=-2}^2 4\pi(5-x)\sqrt{4-x^2} \Delta x$$

$$= 4\pi \int_{-2}^2 (5\sqrt{4-x^2} - x\sqrt{4-x^2}) dx$$

$$= 4\pi \left(5 \times \frac{1}{2} \times \pi \times 2^2 + \frac{1}{3} [(4-x^2)^{3/2}]_{-2}^2 \right)$$

$$= 40\pi^2 \text{ m}^3$$

$$(c) 4x^3 - 12x^2 + 11x - 3 = 0$$

Let roots be $\alpha, d, \alpha, \alpha+d$

$$\text{sum} : 3\alpha = 3 \Rightarrow \alpha = 1$$

$$\Sigma \alpha\beta : 3\alpha^2 - d^2 = \frac{11}{4}$$

$$d^2 = \frac{1}{4}$$

$$d = \pm \frac{1}{2}$$

$\therefore$ roots are $\frac{1}{2}, 1, \frac{1}{2}$

ie solution $x = \frac{1}{2}, 1, \frac{1}{2}$

Question 6

$$(a) 3^n - 1 \geq 2n$$

Show true for $n=1$

$$\text{LHS} = 3^1 - 1 = 2 \quad \text{RHS} = 2 \times 1 = 2$$

Assume true for $n=k$

$$\text{ie } 3^k - 1 \geq 2k$$

Show true for $n=k+1$

$$3^{k+1} - 1 = 3 \times 3^k - 1$$

$$\geq 3 \times (2k+1) - 1$$

$$= 3 \times 2k + 2$$

$$= 2(k+1) + 4k$$

$$\geq 2(k+1) \text{ since } 4k$$

$$\therefore 3^{k+1} - 1 \geq 2(k+1)$$

ie true for $n=k+1$ if true for n
 Since true for $n=1$, also true for
 $n=1+1=2$ and thus true for $n=2$
 and so on for all positive integral

$$(b) (i) \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{-b^2}{a^2} \frac{x}{y}$$

$$\text{at } P \quad m_T = \frac{-b \cos \theta}{a \sin \theta}$$

$$\therefore \text{eq QR} \quad y = \frac{-b \cos \theta}{a \sin \theta} x$$

find Q + R

$$\frac{x^2}{a^2} + \frac{b^2 \cos^2 \theta x^2}{b^2 a^2 \sin^2 \theta} = 1$$

$$\frac{x^2 (\sin^2 \theta + \cos^2 \theta)}{a^2 \sin^2 \theta} = 1$$

$$x^2 = a^2 \sin^2 \theta$$

$$x = \pm a \sin \theta$$

$$x = \pm a \sin \theta, \quad y = \mp b \cos \theta$$

$$\therefore Q + R \quad (\pm a \sin \theta, \mp b \cos \theta)$$

$$(ii) QR = \sqrt{(2a \sin \theta)^2 + (2b \cos \theta)^2}$$

$$= 2 \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta}$$

$\perp$ dist P to QR

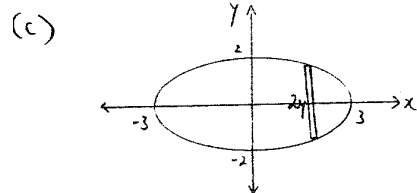
$$d = \frac{1}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}}$$

$$= \frac{ab (\cos^2 \theta + \sin^2 \theta)}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}}$$

$$A = \frac{1}{2} \times 2 \sqrt{a^2 \sin^2 \theta + b^2 \cos^2 \theta} \times \frac{ab}{\sqrt{b^2 \cos^2 \theta + a^2 \sin^2 \theta}}$$

= ab which is independent of θ

∴ Area ΔPQR is independent of position P.



$$\frac{x^2}{9} + \frac{y^2}{4} = 1 \Rightarrow y^2 = \frac{4}{9} (9 - x^2)$$

$$\text{Area cross-section} = 4y^2$$

$$= \frac{16}{9} (9 - x^2)$$

$$\text{Volume slice } \Delta V = \frac{16}{9} (9 - x^2) \Delta x$$

$$\text{Volume } V = \lim_{\Delta x \rightarrow 0} \sum_{x=-3}^3 \frac{16}{9} (9 - x^2) \Delta x$$

$$= \frac{16}{9} \int_{-3}^3 (9 - x^2) dx$$

$$= \frac{32}{9} \left[9x - \frac{x^3}{3} \right]_0^3$$

$$= \frac{32}{9} (27 - 9)$$

$$= 64 \text{ u}^3$$

(d) $W = \frac{z+3}{z} \Rightarrow zw - z = 3$

$$z(w-1) = 3$$

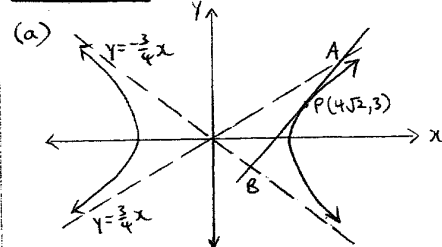
$$z = \frac{3}{w-1}$$

$$|z|=2 \quad \therefore \quad z = |w-1|$$

$$|w-1| = \frac{3}{2}$$

∴ W is a circle, centre (1,0)
radius $\frac{3}{2}$

Question 7



$$\frac{x^2}{16} - \frac{y^2}{9} = 1 \Rightarrow \frac{x}{8} - \frac{2y}{9} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{9x}{16y} \quad m_T = \frac{9 \times 4\sqrt{2}}{16 \times 3} = \frac{3\sqrt{2}}{4}$$

eq. tangent $y-3 = \frac{3\sqrt{2}}{4} (x-4\sqrt{2})$

$$4y - 3\sqrt{2}x + 12 = 0$$

when $y = \frac{3}{4}x$ $3x - 3\sqrt{2}x + 12 = 0$

$$x = \frac{4}{\sqrt{2}-1}$$

pt. A $\left(\frac{4}{\sqrt{2}-1}, \frac{3}{\sqrt{2}-1} \right)$

when $y = -\frac{3}{4}x$ $-3x - 3\sqrt{2}x + 12 = 0$

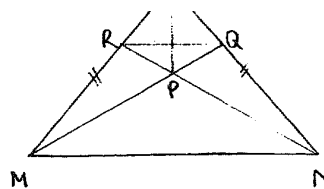
$$x = \frac{4}{1+\sqrt{2}}$$

pt. B $\left(\frac{4}{1+\sqrt{2}}, \frac{-3}{1+\sqrt{2}} \right)$

midpt AB = $\left(\frac{1}{2} \left(\frac{4}{\sqrt{2}-1} + \frac{4}{1+\sqrt{2}} \right), \frac{1}{2} \left(\frac{3}{\sqrt{2}-1} - \frac{3}{1+\sqrt{2}} \right) \right)$

$$= \left(\frac{1}{2} \times 8\sqrt{2}, \frac{1}{2} \times 6 \right)$$

$$= (4\sqrt{2}, 3) \text{ which is P}$$



In ΔMPL and ΔNPL

LM = LN given

LP is common

$\angle PLM = \angle PLN$ P on bisector of $\angle MLN$

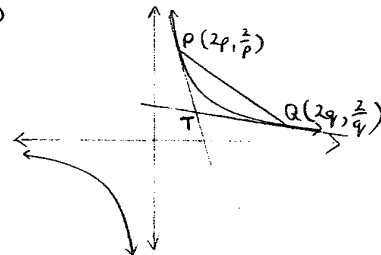
∴ $\Delta MPL \cong \Delta NPL$ SAS

∴ $\angle LMP = \angle LNP$ corresp $\angle s \cong \Delta s$

Interval RQ subtends equal angles $\angle LMP$ and $\angle LNP$ on the same side of it

∴ M, N, Q and R are concyclic

(c)



(i) $m_{PQ} = \frac{\frac{2}{p} - \frac{2}{q}}{2p - 2q}$

$$= \frac{2(q-p)}{pq} \cdot \frac{1}{2(p-q)}$$

$$= -\frac{1}{pq}$$

eq. PQ $y - \frac{2}{p} = -\frac{1}{pq} (x - 2p)$

$$pqy - 2q = -x + 2p$$

$$x + pqy = 2(p+q)$$

tangent at P $x + p^2y = 4p$

(iii) $(2(p+q-1), 2)$ satisfies eq. i

$$2(p+q-1) + 2pq = 2(p+q)$$

$$-2 + 2pq = 0$$

$$pq = 1$$

tangent at P $x + p^2y = 4p$

tangent at Q $x + q^2y = 4q$

solve simult. $(p^2 - q^2)y = 4(p - q)$

$$y = \frac{4(p-q)}{(p-q)(p+q)}$$

$$y = \frac{4}{p+q}$$

$$x + p^2 \times \frac{4}{p+q} = 4p$$

$$x = 4p - \frac{4p^2}{p+q}$$

$$= \frac{4p(p+q) - 4p^2}{p+q}$$

$$= \frac{4pq}{p+q}$$

$$= \frac{4}{p+q} \text{ since } pq = 1$$

∴ $y = x$ is locus of T

Question 8

(a) $(\cos \theta + i \sin \theta)^3$ expand + de M

$$\cos^3 \theta + 3i \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta$$

$$(\cos \theta)^3 = \cos 3\theta + i \sin 3\theta$$

equating real parts

$$\cos 3\theta = \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta)$$

$$= 4 \cos^3 \theta - 3 \cos \theta$$

[Or expand $\cos(2\theta + \theta) \dots$]

$$+ x = \cos \theta$$

$$8x^3 - 6x = -1$$

$$\therefore (4 \cos^3 \theta - 3 \cos \theta) = -1$$

$$2 \cos 3\theta = -1$$

$$\cos 3\theta = -\frac{1}{2}$$

$$3\theta = \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}$$

$$\theta = \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{8\pi}{9}$$

$$\text{solution is } x = \cos \frac{2\pi}{9}, \cos \frac{4\pi}{9}, \cos \frac{8\pi}{9}$$

$$a > b, c > d$$

$$\therefore a - b > 0, c - d > 0$$

$$(a - b)(c - d) > 0$$

$$ac - ad - bc + bd > 0$$

$$ac + bd > ad + bc$$

$$) a \geq c + d, b \geq c + d$$

$$(a - c) \geq d, (b - d) \geq c$$

$$(a - c)(b - d) \geq dc$$

$$ab - ad - bc + dc \geq dc$$

$$ab \geq ad + bc$$

) (i)

$$\text{LHS} = \frac{1 + \sin \theta + i \cos \theta}{1 + \sin \theta - i \cos \theta} \times \frac{1 + \sin \theta + i \cos \theta}{1 + \sin \theta + i \cos \theta}$$

$$= \frac{(1 + \sin \theta + i \cos \theta)^2}{(1 + \sin \theta)^2 - i^2 \cos^2 \theta}$$

$$= \frac{(1 + \sin \theta)^2 + 2i(1 + \sin \theta) \cos \theta - \cos^2 \theta}{(1 + \sin \theta)^2 + \cos^2 \theta}$$

$$= \frac{(1 + \sin \theta)^2 + 2i(1 + \sin \theta) \cos \theta - (1 - \sin^2 \theta)}{(1 + \sin \theta)^2 + (1 - \sin^2 \theta)}$$

$$= \frac{(1 + \sin \theta)^2 + 2i(1 + \sin \theta) \cos \theta - (1 - \sin^2 \theta)(1 + \sin \theta)}{(1 + \sin \theta)^2 + (1 - \sin^2 \theta)(1 + \sin \theta)}$$

$$= \frac{1 + \sin \theta + 2i \cos \theta - 1 + \sin \theta}{1 + \sin \theta + 1 - \sin \theta}$$

$$= \frac{2 \sin \theta + 2i \cos \theta}{2}$$

$$= \sin \theta + i \cos \theta$$

$$(ii) \text{ RHS} = \cos \left(\frac{\pi}{2} - n\theta \right) + i \sin \left(\frac{\pi}{2} - n\theta \right)$$

$$= \sin(n\theta) + i \cos(n\theta)$$

$$= (\sin \theta + i \cos \theta)^n$$

$$= \left(\frac{1 + \sin \theta + i \cos \theta}{1 + \sin \theta - i \cos \theta} \right)^n$$

$$= \text{LHS}$$